RISK 201: Repayment of Debts

Himchan Jeong, FSA

Department of Statistics & Actuarial Science
Simon Fraser University

5.1 Amortization of a debt

- When the **amortization method** is used to repay a loan, the borrower makes **installment payments** to the lender at regular intervals.
- Each installment payment is used:
  - to pay the interest on the outstanding loan balance since the last payment, AND
  - to repay part of the principal borrowed.
- We will be interested in finding the outstanding balance, and the interest/principal portion of each payment.

Amortization schedule

- An **amortization schedule** for a loan identifies the corresponding portions of each payment.
- Dividing each payment into principal and interest is often required in practice. For example, the two may be treated differently for income tax purposes.
- Amortization schedules are easiest to build in a spreadsheet. See the textbook for instructions.

Example 5.1

A \$5000 loan is repaid by 5 equal installments with $i = 10\%$.

Period	Payment (K_t)	Interest (I_t)	Principal (PR_t)	Outstanding balance (OB_t)
0				5,000
1				
2				
3				
4				
5				

General relationships

$$I_t = i \cdot OB_{t-1}$$

$$PR_t = K_t - I_t$$

$$OB_t = OB_{t-1} - PR_t$$

$$L = OB_0 = \sum_{t=1}^{n} PR_t$$

General case (level payments)

Consider a loan to be repaid by n installments of K.

Period	Pmt	$I_t = i \cdot OB_{t-1}$	$PR_t = K - I_t$	OB_t
0				$Ka_{\overline{n}\rceil}$
1	K			
2	K			
...	...	...	...	...
$n-1$	K			
n	K			

Special relationships (level pmts)

$$L = OB_0 = Ka_{\overline{n}|}$$

$$OB_t = Ka_{\overline{n-t}|}$$

$$I_t = i \cdot Ka_{\overline{n-t+1}|} = K\left(1 - v^{n-t+1}\right)$$

$$PR_t = K - I_t = Kv^{n-t+1}$$

Example 5.2 (non-level payments)

A \$10,000 loan is repaid by unequal installments ($i = 10\%$).

Period	Payment (K_t)	Interest (I_t)	Principal (PR_t)	Outstanding balance (OB_t)
0				10,000
1	4,000			
2	3,000			
3	1,500			
4	500			
5	?			

Example 5.3 (level principal repayment)

A \$10,000 loan is repaid by level payments of principal plus interest on the OB at $i = 10\%$.

Period	Payment (K_t)	Interest (I_t)	Principal (PR_t)	Outstanding balance (OB_t)
0				10,000
1			2,000	
2			2,000	
3			2,000	
4			2,000	
5			2,000	

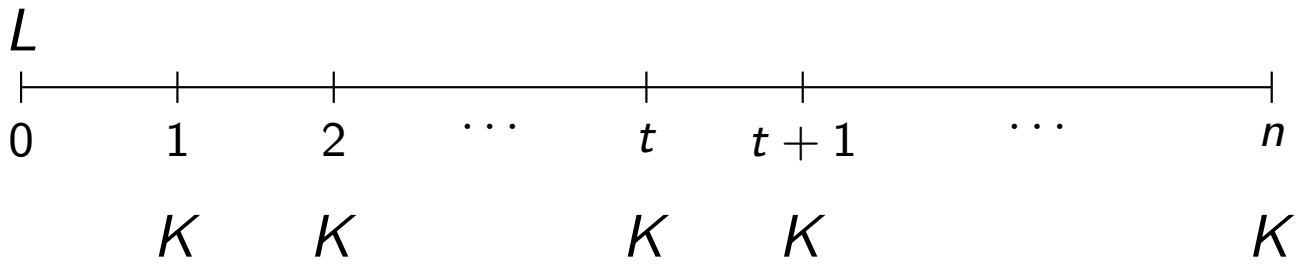
Example 5.4 (non-level rates)

Consider Example 5.2 again. Let the interest rate applicable during period t be i_t.

Period	Payment (K_t)	Rate (i_t)	Interest (I_t)	Principal (PR_t)	Outstanding balance (OB_t)
0					10,000
1	2,500	10%			
2	2,500	8%			
3	2,500	10%			
4	2,500	5%			
5		7%			

5.2 Outstanding balance

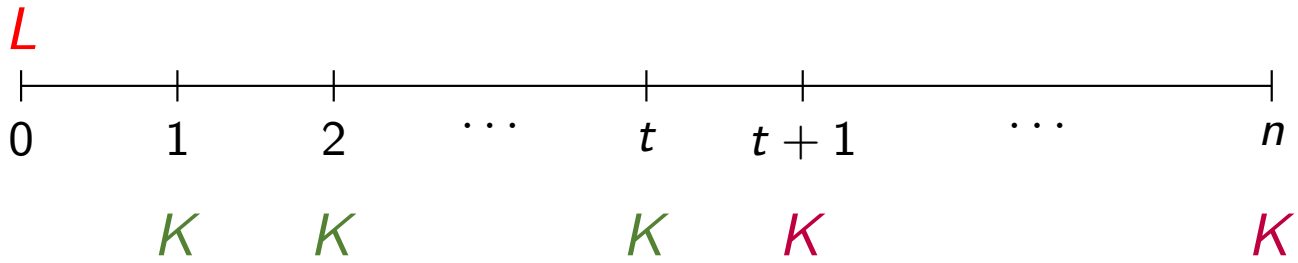
Consider a loan L that is amortized by n level installment payments of K.



Focal date at time 0: $L = Ka_{\overline{n}|i}$

Focal date at time t: $Ka_{\overline{n-t}|i}$

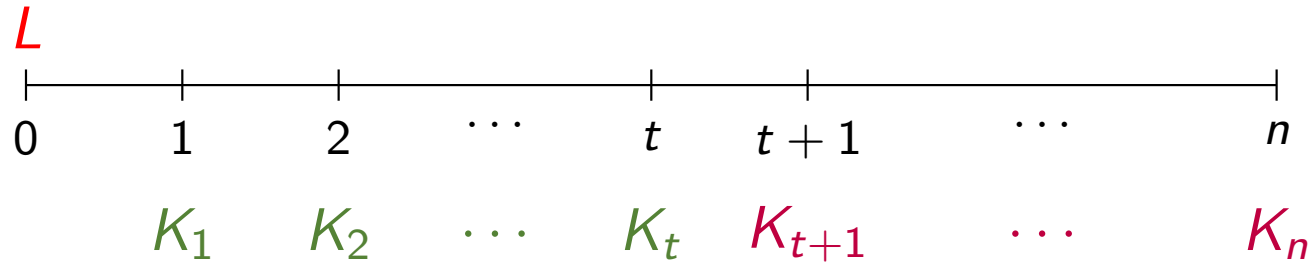
Outstanding balance (level pmts)



Prospectively: $OB_t = Ka_{\overline{n-t}|i}$

Retrospectively: $OB_t = L(1+i)^t - Ks_{\overline{t}|i}$

Outstanding balance (general)



$$L = \sum_{s=1}^{n} K_s v^s$$

$$OB_t = \sum_{s=t+1}^{n} K_s v^{s-t} = L(1+i)^t - \sum_{s=1}^{t} K_s (1+i)^{t-s}$$

Example 5.1 revisited

Period	K_t	I_t	PR_t	OB_t
0				5,000.00
1	1318.9874	500	818.9874	4181.012596
2	1318.9874	418.1013	900.8861	3280.126452
3	1318.9874	328.0126	990.9748	2289.151693
4	1318.9874	228.9152	1090.072	1199.079458
5	1318.9874	119.9079	1199.079	0.00

$$OB_2 = 5000(1.1)^2 - 1318.9874\, s_{\overline{2}|10\%} \approx 1318.9874\, a_{\overline{3}|10\%}$$

Example 5.1 revisited

Period	K_t	I_t	PR_t	OB_t
0				5,000.00
1	1318.9874	500	818.9874	4181.012596
2	1318.9874	418.1013	900.8861	3280.126452
3	1318.9874	328.0126	990.9748	2289.151693
4	1318.9874	228.9152	1090.072	1199.079458
5	1318.9874	119.9079	1199.079	0.00

$$OB_2 = 1318.9874\, a_{\overline{3}|10\%}$$

Example 5.5

A 500,000 mortgage is taken out over 25 years. The interest rate is fixed at $i^{(12)} = 3\%$ for 5 years.

a) Calculate the monthly payment.

$$500000 = Ka_{\overline{300}|0.0025}, \qquad K = 2371.057$$

b) What is the outstanding balance after 5 years?

$$Ka_{\overline{240}|0.0025} = 427572.4$$

c) How much interest is paid over the first 5 years?
Total payments – paid principal =

$$60K - \left(Ka_{\overline{300}|0.0025} - Ka_{\overline{240}|0.0025}\right) = 69790.77$$

5.3 Refinancing a loan

- Mortgage loans are long-term commitments, but even on a fixed-rate mortgage the interest rate is only guaranteed for a portion of the amortization period.

- When rates change, it may be advantageous for the borrower to refinance the loan. This means getting a new loan that pays off the outstanding balance on the original loan and establishes new periodic payments.

Penalties

- Borrowers can usually refinance a mortgage **at the end of the term** without penalty.
- If a borrower wishes to refinance a mortgage **during the term,** there may be a penalty to compensate the lender for lost interest.
- A common penalty is 3 times the monthly interest on the outstanding balance.
- Borrowers must take this penalty into account when deciding whether to refinance.

Example 5.6

Consider the mortgage in Example 5.5. After 2 years, the 5-year fixed rate available in the market is $i^{(12)} = 2.4\%$. The borrower considers refinancing at this rate.

a) Calculate the outstanding balance on the original mortgage after 2 years.

$$K a_{\overline{276}|0.0025} = 472306.7$$

b) What would be the new mortgage payments if refinancing the loan (amortizing over 23 years)?

$$K' a_{\overline{276}|0.002} = 472306.7, \qquad K' = 2228.46$$

Example 5.6, continued

c) What is the outstanding balance at the end of the first 5 years if the mortgage is refinanced?

$$K' a_{\overline{240}|0.002} = 424433.2$$

Example 5.6, continued

d) How much interest is paid over the first 5 years if the mortgage is refinanced?

Total payments – paid principal =

$$24K + 36K' - \left(500{,}000 - K' a_{\overline{240}|0.002}\right)$$

$$= 61563.26$$

Example 5.6, continued

e) The lender charges 3 times the monthly interest on the outstanding balance as a penalty. How much is it?

$$3 \times 0.25\% \times K a_{\overline{276}|0.0025} = 3542.301$$

f) Is it worthwhile for the borrower to refinance at the lower rate?

Note that you are paying $K - K'$ less for the remaining 276 months if you refinance. Therefore, PV of refinance at time 2:

$$= (K - K')\, a_{\overline{276}|0.002} - 3542.301 = 27221.42$$

Thus, it is worthwhile to refinance at the lower rate.

5.4 Sinking funds

- A **sinking fund** is a side fund used to save up for a significant financial outlay in the future, such as an equipment purchase or replacement, a major renovation or other capital investment, etc.
- Deposits of K are made to the sinking fund, which earns a rate j per period and accumulates to the desired amount A after n periods:

$$K s_{\overline{n}|j} = A$$

Example 5.7

(Brown & Kopp 5.4.A2) A company wants to save \$100,000 over the next 5 years to expand its plant facility. How much must be deposited at the end of each year if the deposits earn interest at $i^{(1)} = 8\%$?

$$100{,}000 = K s_{\overline{5}|0.08}, \qquad K = 17045.65$$

Sinking fund schedule (savings)

Time	Interest on SF	Deposit (EOY)	Amount in fund
1	–	17,045.65	17,045.65
2	1,363.65	17,045.65	
3		17,045.65	
4		17,045.65	
5		17,045.65	
t		$K = A/s_{\overline{n}\rceil j}$	$Ks_{\overline{t}\rceil j}$

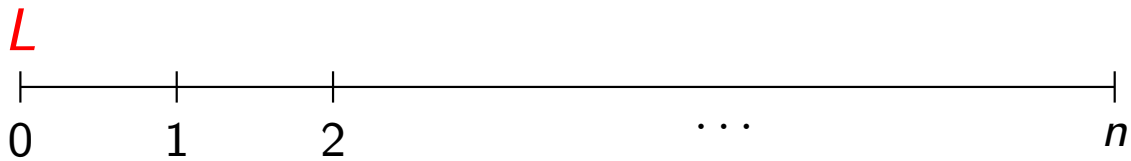
SF method of loan repayment

- Under the **amortization method**, the borrower makes periodic installment payments to the lender which cover **both principal and interest**.
- An alternative is to **make only interest payments during the term** of the loan and **repay the original principal in one lump sum** at the end. If the borrower "saves up" for the principal payment in a sinking fund, this is known as the **sinking fund method** of repaying a loan.

SF method: key interest rates

- The lender charges interest at periodic rate i. We call this the lending rate.
  - The lending rate on a SF-type loan may or may not be the same as the interest rate charged on an amortization-type loan.
- The sinking fund pays interest on the deposits at periodic interest rate j. We call this the savings rate.
  - This savings rate is usually less than the lending rate ($j < i$).

Cash flows under SF method



To lender:	iL	iL		$iL + L$
To SF:	K	K	$\cdots$	K

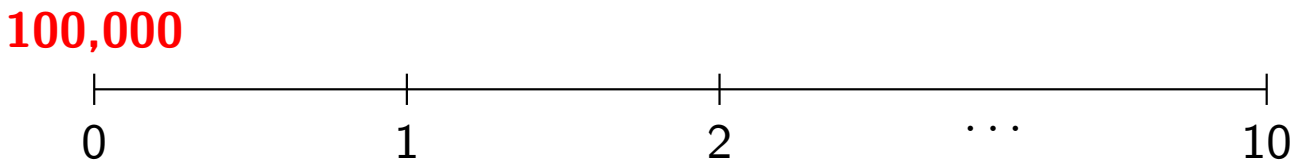
where $L = K s_{\overline{n}|j}$

Total cost of debt to borrower: $iL + \dfrac{L}{s_{\overline{n}|j}}$

Example 5.8

- A loan of 100,000 is to be repaid by 10 annual payments beginning one year after issue.
- The lender wants annual payments of interest at a rate of 10% and repayment of the principal in a single lump sum at the end of 10 years.
- The borrower can accumulate the principal in a sinking fund earning an annual interest rate of 8%, and decides to do this by means of 10 level deposits starting one year after the loan is made.

Example 5.8: cash flows



To lender:

To SF:

Example 5.8: sinking fund schedule

	Original loan		Sinking fund			Periodic cost	Book value
t	Interest	OB	Deposit	Interest	AV	of debt	of loan
1	10,000	100,000	6,902.95	0	6,902.95	16,902.95	93,097.05
2	10,000	100,000	6,902.95	552.24	14,358.13	16,902.95	85,641.87
..	...	...	...	...	...	...	...
9	10,000	100,000	6,902.95	5,873.93	86,200.97	16,902.95	13,799.03
10	10,000	100,000	6,902.95	6,896.08	100,000	16,902.95	0

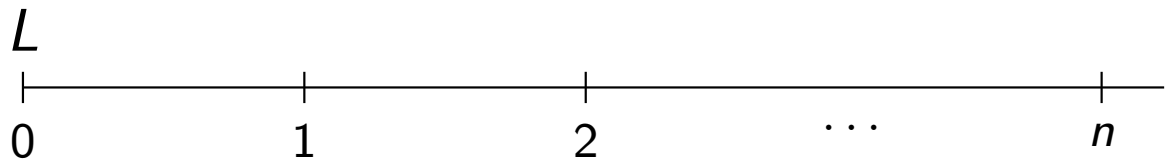
General sinking fund schedule

	Original loan		Sinking fund			Periodic cost	Book value
t	**Int**	**OB**	**Deposit**	**Interest**	**AV**	**of debt**	**of loan**
1	iL	L	K	0	K	$iL + K$	$L - K$
2	iL	L	K	jK	$Ks_{\overline{2}\rceil j}$	$iL + K$	$L - Ks_{\overline{2}\rceil j}$
..	...	...	...	...	...	...	...
$n-1$	iL	L	K	$jKs_{\overline{n-2}\rceil j}$	$Ks_{\overline{n-1}\rceil j}$	$iL + K$	$L - Ks_{\overline{n-1}\rceil j}$
n	iL	L	K	$jKs_{\overline{n-1}\rceil j}$	$Ks_{\overline{n}\rceil j}$	$iL + K$	0

5.5 Comparison of amortization and sinking-fund methods

- When borrowers decide whether to use the amortization or sinking fund method of loan repayment, they should **compare the periodic payments** made under both methods.
- These will depend on:
  - the loan rate charged under the amortization method (i_A)
  - the loan rate charged under the SF method (i_S)
  - the savings rate earned on SF deposits (j)

Cash flow comparison



Amortization method: $\frac{L}{a_{\overline{n}|i_A}} \quad \frac{L}{a_{\overline{n}|i_A}} \quad \cdots \quad \frac{L}{a_{\overline{n}|i_A}}$

Sinking-fund method: $i_S L + \frac{L}{s_{\overline{n}|j}} \quad i_S L + \frac{L}{s_{\overline{n}|j}} \quad \cdots \quad i_S L + \frac{L}{s_{\overline{n}|j}}$

Cash flow comparison, continued

When $i_A = i_S = j$:

- Periodic payment under amortization method: $\dfrac{L}{a_{\overline{n}|j}}$
- Periodic payment under SF method:

$$jL + \frac{L}{s_{\overline{n}|j}} = L\left(j + \frac{j}{(1+j)^n - 1}\right) = L\left(\frac{j(1+j)^n - j + j}{(1+j)^n - 1}\right)$$
$$= L\left(\frac{j}{1 - v^n}\right) = \frac{L}{a_{\overline{n}|j}}$$

In this case, the two methods are equivalent

Cash flow comparison, continued

When $i_A = i_S$ and $i_S > j$:

- Periodic payment under amortization method: $\dfrac{L}{a_{\overline{n}|i_S}}$
- Periodic payment under SF method:

$$i_S L + \frac{L}{s_{\overline{n}|j}} > i_S L + \frac{L}{s_{\overline{n}|i_S}} = \frac{L}{a_{\overline{n}|i_S}}$$

Amortization method is better for the borrower.

Cash flow comparison, continued

When $i_A = i_S$ and $i_S < j$:

- Periodic payment under amortization method: $\dfrac{L}{a_{\overline{n}|i_S}}$
- Periodic payment under SF method:

$$i_S L + \frac{L}{s_{\overline{n}|j}} < i_S L + \frac{L}{s_{\overline{n}|i_S}} = \frac{L}{a_{\overline{n}|i_S}}$$

Sinking fund method is better for the borrower.

Other considerations

A borrower may still choose the sinking fund method even if the periodic cost of payment is higher than under the amortization method because:

- The borrower controls the SF deposit amount. If it needs to reduce payments for a while, it can catch up later.
- The borrower can take risk in investing the deposits and maybe earn a higher return.
- The borrower can usually negotiate a lower loan rate than with amortization.

Example 5.9

A loan of 100,000 is to be repaid by 10 payments beginning one year after the loan is made.

- If the borrower chooses the amortization method, the lender charges a loan rate of 11%.
- If the borrower chooses to pay interest only on the loan, the lender is willing to reduce the loan rate to 10%. The borrower can get a savings rate of 8%.

Which one should the borrower choose?

Example 5.9, continued

General equivalence

- Suppose a loan is repaid using the sinking fund method, with loan rate i_S and savings rate j.
- Let i' be the loan rate under the amortization method that makes the periodic cost of debt equal under the two methods:

$$i_S L + \frac{L}{s_{\overline{n}|j}} = \frac{L}{a_{\overline{n}|i'}}$$

- Approximate this loan rate as $i' = i_S + \dfrac{i_S - j}{2}$.